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Final Report

Origins of Mars's Moons: Physics-Structured ML Using Asteroid Tidal-Disruption SPH Simulations

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Abstract

One theory for the formation of Mars's moons, Phobos and Deimos, is that close encounters between asteroids and Mars can tidally disrupt the asteroid and leave some debris gravitationally bound to the planet. This process is thoroughly modelled by smoothed-particle hydrodynamics (SPH), but testing every possible combination of inputs is computationally costly. In this study, bound mass fraction (BMF) is predicted from the scenario inputs using machine learning models after SPH outputs are converted into post-processed outcome measurements. The primary model, a tuned Gradient Boosting regressor using the original scenario inputs achieves a grouped cross-validated R^2 of 0.9328, mean absolute error (MAE) of 0.0277, and root mean squared error (RMSE) of 0.0400. Adding the tested physics-derived features does not improve the tuned Gradient Boosting model, so the simpler tuned raw-input model is retained. The models are assessed using grouped cross-validation. In a controlled set, predicted BMF generally decreases from roughly 0.48 at 1.2 Mars radii to roughly 0.12 at 3.0 Mars radii. A similarly clear decline occurs as encounter speed increases, from roughly 0.45 at low velocity to near zero by about 1.2-1.4 km/s. Overall, velocity has the strongest model dependence, followed by periapsis and spin, although their relative importance varies with encounter conditions. Matched comparisons further show that the effects of spin and velocity vary with the encounter conditions. Despite falling within the numerical training ranges, held-out errors reach up to 7.2 percentage points at periapsis 1.4 Mars radii in a sparsely sampled region, demonstrating that being within the input range does not ensure a trustworthy prediction. Therefore, the model does not replace SPH or prove physical causality, but it is helpful for quickly screening well-supported encounter scenarios and highlighting parameter relationships that call for additional SPH research.

1. Introduction

1.1 Research Problem and Questions

The origin of Phobos and Deimos, the moons of Mars, is still a mystery. A close asteroid contact with Mars is one suggested route. During a sufficiently close encounter, differences in Mars's gravitational pull across the asteroid can produce strong tidal forces that deform or disrupt it. While some debris might escape, others might stay gravitationally attached to Mars. This retained material may subsequently undergo orbital evolution and potentially accrete into moons (Kegerreis et al., 2024).

Smoothed-particle hydrodynamics (SPH) works well in this case because its particles move with the material as the material deforms, fragments, and interacts gravitationally. However, simulating each new arrangement is computationally expensive. It is simply not practical to test every possible combination of asteroid mass, closest approach, encounter speed, spin, and numerical simulation settings. Using the present simulation archive, this study creates a machine-learning (ML) surrogate to screen potential scenarios and identify whether new SPH simulations are necessary.

There are three research questions addressed here: (1) how do orbital and physical parameters affect disruption and bound retention; (2) which inputs matter most for retained debris; and (3) can machine-learning models predict outcomes within the sampled data while showing when a prediction should or should not be trusted?

1.2 Significance and Originality

The work has both scientific and computational significance. From a scientific perspective, simulations of a mechanism that might provide debris to the Martian system can be found in the SPH archive. It solves a frequent computational issue in simulation-based science: while a thorough solver can accurately simulate one setup, it is computationally costly to run multiple setups. Full SPH simulations can be concentrated on situations where ML models are most helpful because, once trained, they can swiftly analyse many combinations.

The primary contribution of this project is a workflow that combines physical interpretation with explicit tests of ML prediction reliability. Each simulation's particle-level SPH outputs are combined into a single set of results by the workflow, which also tests the model on physical simulations that weren't used for training, compares matched and controlled cases, calculates the amount of similar SPH data that is present nearby, and uses these checks to suggest whether a new SPH run is necessary.

1.3 Literature Review and Research Gaps

The formation of Mars' moons, Phobos and Deimos, has been explained in several ways. Debris disks surrounding Mars can be created by giant-impact simulations (Citron et al., 2015; Canup & Salmon, 2018). Phobos and Deimos may originate from a disrupted shared parent body, according to orbital studies, and temporary inner moons may influence how material in the outer disk accretes (Rosenblatt et al., 2016; Bagheri et al., 2021). Nevertheless, little is known about the mechanisms that create and hold onto debris in the vicinity of Mars. So, by demonstrating that a close asteroid contact with Mars can tidally destroy the asteroid while trapping part of its debris, Kegerreis et al. (2024) offer a significant alternative creation pathway. The current project is directly motivated by this. Although their simulations demonstrate significant amounts of bound debris across various periapsis, velocity, mass, and spin settings, each new configuration necessitates a new simulation. This study investigates whether an ML model that screens new configurations prior to SPH execution can be trained using the current archive.

Since SPH is used to simulate these tidal-disruption scenarios, it is also crucial to comprehend how the simulation data are generated and handled. Because SPH is a mesh-free Lagrangian approach, computational particles follow the motion of the material they represent (Price, 2012). It is therefore appropriate for significant deformation and fragmentation. However, the subsequent processing of the simulation also affects the measured pieces. Here, post-processing techniques like friends-of-friends (FoF) use a selected connecting length to group neighbouring particles; therefore, the reported number and mass of fragments depend in part on the encounter physics as well as that processing option (More et al., 2011). Bound mass is handled differently: a fragment's orbital energy in relation to Mars determines whether it is bound or unbound.

The particle-level SPH outputs are transformed into scalar outcomes through post-processing, creating a compact tabular dataset appropriate for ML. Because tree-based models perform well with tiny, nonlinear tabular datasets, this makes the problem a good fit for machine learning (Grinsztajn et al., 2022). In the past, outcomes from comparable SPH collision and giant-impact simulations have also been predicted using tree-based ML surrogate models (Cambioni et al., 2019; Timpe et al., 2020). They are still not widely used for asteroid tidal-disruption encounters, though.

2. Methodology

2.1 Data Archive, Inputs, and Post-Processing

489 SPH outputs totalling over 218 GB of Hierarchical Data Format version 5 (HDF5) particle data are included in the collection. These particle outputs are transformed into fragment and bound-mass measurements through post-processing. The physical scenario inputs and numerical simulation settings are provided by the SPH setup, while bound-mass measurements are calculated during post-processing. The main outcome/model target in this project is the Bound Mass Fraction (BMF), which is the fraction of the original asteroid mass that remains gravitationally bound to Mars after the encounter.

*Table 1. Main SPH setup and post-processing parameters.
Some parameter values appear much more often than others in the archive.*

Parameter	Coverage range	What it means
Asteroid mass	10^{18} to 10^{20} kg; 10^{20} dominates	Mass of the incoming asteroid. Its effect on BMF also depends on the other encounter conditions.
Periapsis	1.1 to 3.0 Mars radii; 1.2 and 1.6 Mars radii common	Closest distance between the asteroid and Mars. Smaller values mean a closer encounter and stronger tidal forces.
Encounter speed	0 to 3.0 km/s; $v_{\infty}=0$ km/s dominates	Asteroid speed far from Mars before the encounter. Higher encounter speeds generally reduce the time available for tidal interaction and capture.

Spin	No spin; equatorial x- or y-axis spin; prograde z-axis spin; retrograde z-axis spin; unevenly sampled	Rotation state of the asteroid. Spin can alter the disruption and bound-retention outcome, and the available spin states are unevenly sampled.
Resolution	Approximately 10^5 to 10^7 particles; $10^{6.5}$ particles dominates	Number of SPH particles used in the simulation. This is a numerical simulation setting rather than a physical property of the encounter.
FoF linking length (<i>post-processing</i>)	0.0001-0.0126; 0.004 common	Maximum separation used by the friends-of-friends algorithm to assign nearby particles to the same fragment. It can change the measured fragment properties but does not change the physical encounter.

How the post-processing works is that FoF first groups nearby debris particles into resolved fragments, whose orbital properties can be analysed using their centre-of-mass positions and velocities. The bound-mass calculation additionally includes debris that is not assigned to a resolved FoF fragment, and BMF is measured relative to the original asteroid mass.

2.2 Prediction Model Choices and Rationale

The primary ML task is to predict BMF, the proportion of the initial asteroid mass expected to remain bound to Mars. This preserves the distinction between low, medium, and high retained fractions. Furthermore, because Kegerreis et al. (2024) found that “tens of percent of an unbound asteroid’s mass can be captured and survive beyond collisional timescales”, a 10% BMF threshold is used as a simple screening criterion for higher-retention cases. This threshold is not a physical limit for moon formation, but is used only for classification and prioritisation.

For the prediction tasks, several baseline models were used. Logistic Regression models the probability of a case belonging to one of two classes from the input features, so it is used as a simple baseline for the $\text{BMF} \geq 10\%$ classification task (Knauer et al., 2024). Random Forest combines many decision trees fitted to different samples of the data and averages their predictions, allowing it to capture nonlinear relationships while reducing sensitivity to individual cases

(Breiman, 2001; Cambioni et al., 2019). Gradient Boosting also combines decision trees, but builds them sequentially so that each new tree reduces errors left by the previous trees. This can capture more detailed changes between encounter settings (Friedman, 2001; Valencia et al., 2019; Timpe et al., 2020).

To determine whether prediction performance could be improved beyond the baseline models, Gradient Boosting hyperparameters were tuned and physics-derived attributes calculated from the original scenario inputs were tested as a separate feature-engineering experiment. These included squared encounter speed, inverse periapsis, spin frequency, asteroid radius, encounter eccentricity, periapsis-speed interaction, and estimated time spent within selected Mars-centred radii. The full hyperparameters and definitions of the physics-derived attributes are provided in Appendix A, Tables A1-A2.

2.3 Model Testing and Performance Metrics

One of the main issues with the archive is that it contains repeated rows from the same physical SPH simulation, for example when the same simulation is post-processed using different FoF linking lengths. With a random row split, the model could therefore be trained on one processed version of a simulation and tested on another version of the same simulation. To prevent this, 5-fold GroupKFold cross-validation was used. GroupKFold divides the data into five folds while keeping all rows from the same physical simulation in the same fold. Each out-of-fold (OOF) prediction is therefore produced by a model that was not trained on any row from that physical simulation.

Regression performance is measured using R^2 , mean absolute error (MAE), and root mean squared error (RMSE). Classification performance is measured using balanced accuracy, F1 score, and receiver operating characteristic area under the curve (ROC AUC), which measures how well the model separates the two classes across classification thresholds.

2.4 How the Physical Effects Were Analysed

Although overall scores are helpful, they do not demonstrate whether the model exhibits reasonable physical behaviour in various scenarios. Three complementary analyses are employed to verify this. To determine how heavily the model depends on each physical input across the dataset, an input-removal test, also known as ablation, first eliminates one original input from the model and analyses how much model performance declines (Section 3.4). Second, to illustrate the direction and form of the model response, controlled tests alter one input while maintaining the others constant (Section 3.5). Third, matched SPH comparisons examine whether an input's significance varies depending on encounter circumstances (Section 3.5).

2.5 When to Trust the Model

The method also determines whether a new setup is comparable to SPH simulations that are already in the archive because model accuracy alone cannot demonstrate the reliability of each individual forecast because an overall score can conceal areas where the model has insufficient data or produces larger errors. When there are multiple independent SPH simulations and a low held-out prediction error in the surrounding area, predictions are more reliable. When the input combination is outside the sampled region, close to the edge of the available data, or poorly sampled, predictions are deemed less accurate. These reliability checks are analysed in Section 3.6.

The final decision then separates two questions: (1) how strongly the asteroid breaks up, and (2) how much material remains bound to Mars. Breakup is described by the largest-remnant fraction $f_{LR} = M_{largest}/M_{parent}$ where score above 0.90 is mostly intact, 0.50-0.90 is partial breakup, and at or below 0.50 is severe disruption. Retention is measured separately with BMF and Mars-bound mass, $M_{bound} = BMF \times M_{parent}$, this is important because severe breakup can still leave almost no material bound to Mars (Richardson et al., 1998; Walsh & Richardson, 2006).

3. Results

3.1 Exploratory Physical Trends

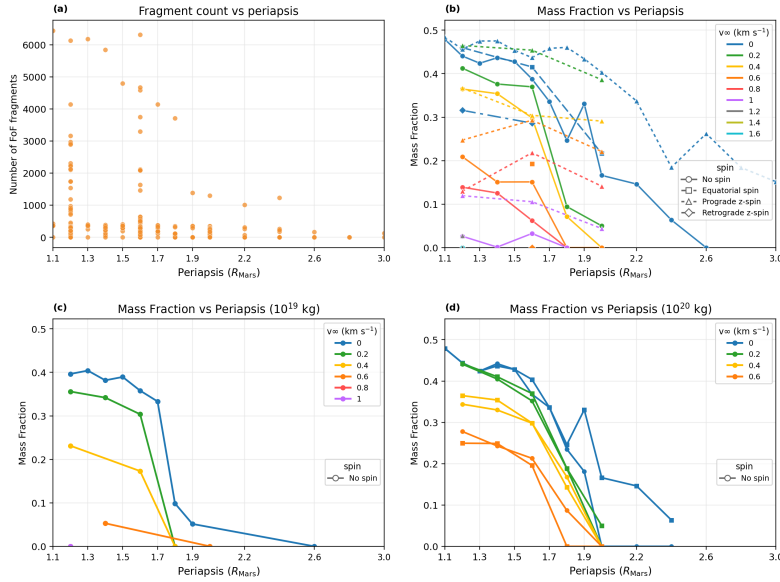


Figure 1. Exploratory SPH trends and mass-dependent bound-retention regimes. (a) Fragment count versus periapsis. (b) BMF across all sampled encounter conditions. (c-d) BMF results for the 10^{19} and 10^{20} kg asteroid-mass groups, showing that the combined scatter includes distinct retention regimes.

Although close interactions do not always result in strong fragmentation or high BMF, Figure 1 illustrates that both fragmentation and bound retention typically decrease as periapsis increases. The mean BMF rises with tiny periapsis and falls with increasing encounter speed. The relative impacts also vary throughout the sampling range: separation between spin states becomes more apparent around 1.7-2.0 Mars radii, whereas velocity differences are more noticeable at roughly 1.1-1.6 Mars radii. Some prograde-spin cases retain non-zero BMF to larger periapsis, although these regions are more sparsely sampled.

The combined BMF graph is highly irregular because it contains simulations from different asteroid-mass regimes. Separating the mass cases shows that the apparent scatter reflects regime-dependent bound mass rather than a single periapsis trend. The mass-separated panels also show a strong asymmetry in the sampled archive: the 10^{20} kg cases are much more heavily represented (271 cases) than the 10^{19} kg cases (81 cases), and they span substantially higher BMF and remain non-zero over a wider periapsis range. The 10^{19} kg cases generally reach lower BMF than 10^{20} kg cases across the sampled combinations. However, because the two mass groups do not contain the same combinations of velocity, spin, and periapsis, this should not be interpreted as a direct causal mass effect. Overall, the results show regime-dependent behaviour, where distinct bound-mass outcomes result from varying relative effects of parameters across the parameter space rather than a single parameter acting independently.

3.2 Eccentricity and Bound Retention

It was previously demonstrated that fragmentation and BMF do not follow a single, straightforward input trend, thus I want to see if relevant physical parameters can more accurately characterize this behaviour. To provide a single physically significant measure of the orbit shape and enable retention to be analysed using the entire encounter trajectory rather than as distinct factors, periapsis and encounter speed are combined to form an encounter eccentricity.

The outcome demonstrates that retention is concentrated close to the parabolic limit, where eccentricity is close to 1, and generally decreases as the encounter becomes more hyperbolic, as Figure 2 illustrates. Faster, more hyperbolic fly-bys leave less material bound to Mars because they spend less time in capture-friendly conditions.

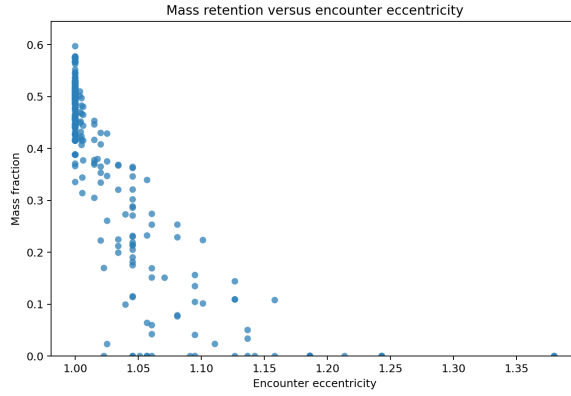


Figure 2. Bound mass fraction versus encounter eccentricity.
Strong and non-zero retention quickly disappears as the encounter becomes more hyperbolic.

3.3 Model Performance and Screening Use

After examining the physical trends in the SPH archive, the next step is to test how accurately these outcomes can be predicted from the scenario inputs. R^2 measures how much of the variation in BMF is reproduced by the model, while MAE and RMSE measure the typical prediction error, with lower values indicating better predictions.

The baseline model comparisons support Gradient Boosting for continuous BMF prediction. Random Forest produces $R^2 = 0.9191$, MAE = 0.0293, and RMSE = 0.0439, while Gradient Boosting achieves $R^2 = 0.9262$, MAE = 0.0296, and RMSE = 0.0419. Gradient Boosting is therefore the stronger baseline model for this dataset.

For the $\text{BMF} \geq 10\%$ screening task, Logistic Regression obtains aggregated balanced accuracy of 0.849 and ROC AUC of 0.925, indicating that the two retention groups are reasonably separable. Continuous BMF continues to be the main output; the threshold is simply utilized for prioritisation.

Tuning Gradient Boosting improves performance to $R^2 = 0.9328$, MAE = 0.0277, RMSE = 0.0400 (Appendix A, Tables A1-A2). Adding the tested physics-derived features does not improve the tuned Gradient Boosting model, so the model using the original scenario inputs is retained.

3.4 Overall Physical Parameter Importance

The input-removal test yields the most understandable overall result. The amount of predictive information lost is indicated by the decrease in R^2 when each original physical input is removed from the tuned Gradient Boosting model. Removing velocity causes the biggest drop ($\Delta R^2 = -0.508$). Removing periapsis gives $\Delta R^2 = -0.466$, and removing spin gives $\Delta R^2 = -0.246$. These

Commented [KN1]: Overall Parameter Importance
Removing periapsis: $\Delta R^2 = -0.574$
Removing velocity: $\Delta R^2 = -0.296$
Removing spin: $\Delta R^2 = -0.316$

Overall Rule
periapsis > spin > velocity

values are separate ablation experiments and should not be added together. They display the degree to which the model depends on each input, not whether increasing that input directly raises or lowers BMF. Velocity has the strongest overall model dependence, followed by periapsis and spin, although this order may vary locally.

A different response is required for fragmentation. The most extreme fragmentation in the SPH data requires very small periapsis, yet a close encounter does not ensure significant breakdown. FoF connecting length has a direct impact on fragment count as well. This analysis does not provide a single, straightforward global ranking for fragmentation because of this. While the actual fragment pattern depends on the entire encounter and how fragments are gathered during post-processing, periapsis indicates where the largest disruption may occur.

3.5 Checking Physical Trends and Condition-Dependent Effects

The overall ranking of the inputs on which the model is most dependent does not demonstrate how BMF varies with each input or whether an input's significance stays constant under various encounter circumstances. Two follow-up issues are posed in this section: (1) how does BMF change when one input is changed while the others remain unchanged, and (2) do the impacts of significant inputs hold true under various encounter conditions?

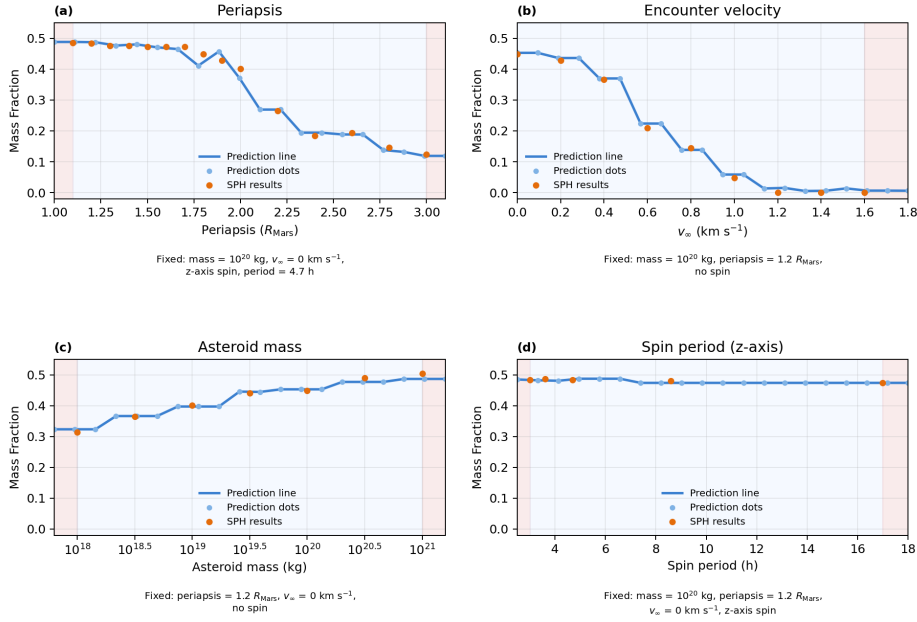


Figure 3. BMF checks where one input changes at a time. Dots are available SPH simulation results and curves show tuned Gradient Boosting predictions with the other inputs fixed. Blue background shows the interpolation range, while red background shows the extrapolation range.

Figure 3 compares controlled profiles in which one scenario input is varied while the others are fixed to a representative, well-supported combination. The fixed conditions may differ between panels, so the profiles show conditional responses rather than directly comparable effect sizes.

Predicted BMF decreases substantially with increasing periapsis, from roughly 0.48 at low periapsis to about 0.12 at 3 Mars radii in this controlled scenario. BMF also decreases with encounter velocity, from about 0.45 at low velocity to near zero by about 1.2-1.4 km/s.

For asteroid mass, seven matched physical scenarios show an overall increase in BMF with mass, although the Gradient Boosting response is stepwise and should not be interpreted as a physical threshold. Spin period has a smaller effect in the selected z-axis-spin scenario, showing that an input's influence depends on the other encounter conditions.

Far outside the sampled parameter ranges, tree-based predictions become stepwise or constant and need not represent physical trends (Appendix Figure A1).

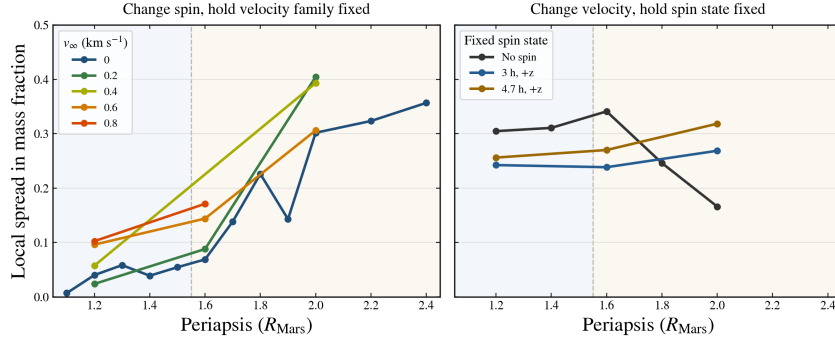


Figure 4. BMF differences within the matched 10^{20} kg. Left: BMF range across spin states with velocity fixed. Right: BMF range across velocities with spin fixed. These ranges show how strongly outcomes differ within this subset; they do not show direction or prove causation.

Because the archive contains matching situations where these parameters can be changed while mass and other numerical and post-processing settings remain unchanged, spin and velocity are used in Figure 4. Next, periapsis is changed throughout the comparison. When velocity is fixed, the difference between the largest and smallest BMF over all possible spin states is represented by $\Delta\text{BMF}_{\text{spin}}$ in the left panel. In the right panel, spin is fixed and $\Delta\text{BMF}_{\text{velocity}}$ falls within the same range across all possible velocities. A higher ΔBMF indicates that, in certain circumstances, altering that parameter has a greater impact on the retained mass.

Figure 4 shows that the magnitude of both spin and velocity effects changes with periapsis and the surrounding encounter conditions. Neither parameter has a constant effect across the sampled range: spin produces larger BMF differences in some regimes/predominates under some situations, while velocity produces larger differences in others. These findings support the importance of regime-dependent parameters, so the overall ablation ranking should not be interpreted as universal.

3.6 Where the Model Can and Cannot Be Trusted

The quantity of local SPH data and prediction error both influence trust. Instead of counting repeated FoF rows, the coverage-error maps count distinct physical simulations and compare that number to the average held-out absolute error in the same area. While sparse or fringe regions can have significantly worse error, regions with many simulations typically have lower error. Low error is insufficient on its own; certain poorly sampled locations appear simple simply because the true BMF is typically zero.

Because the model depends on input combinations, a case may fall inside the range of each individual input and yet have inadequate support. Thus, held-out error and pairwise parameter

coverage are compared with local simulation coverage in Figure 5. Individual input ranges conceal unsampled combinations, which are revealed by pairwise mapping. Only when comparable SPH cases are close by in the input space is a prediction regarded as interpolation.

Coverage, held-out error, and observed mass fraction across the sampled parameter space

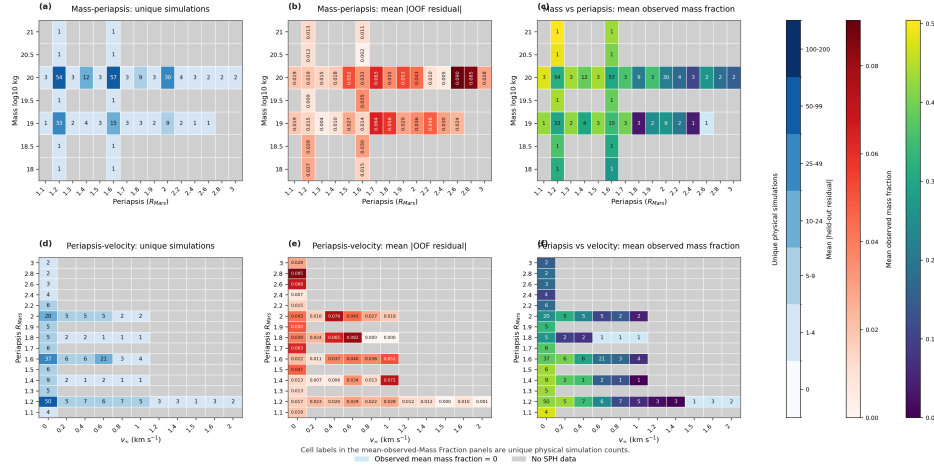


Figure 5. Coverage, held-out error, and observed BMF across the sampled parameter space. Panels (a,d) show unique physical SPH simulation counts, panels (b,e) show mean held-out absolute error, and panels (c,f) show mean observed BMF with unique simulation counts overlaid. Grey indicates no SPH data. In the observed-BMF panels, light blue indicates mean observed BMF = 0, while other colours indicate mean observed BMF > 0.

A specific example of this issue can be seen from this two sparsely sampled periapsis-velocity cells in Table 2. At a fixed periapsis of 1.4 Mars radii, the $v_{\infty} = 0.8$ and 1.0 km/s cells each contain only one unique physical SPH simulation (see Figure 5). The table compares the observed BMF, held-out prediction, and absolute error.

Table 2. Sparse failure case at periapsis 1.4 Mars radii: both velocity cells are within the sampled range but contain only one unique physical SPH simulation.

Velocity	Actual BMF	OOF prediction	Abs. error	Failure
0.8	0.150849	0.103632	0.0472	Underpredicts
1.0	0.023327	0.095684	0.0724	Overpredicts

The held-out model underpredicts the $v_{\infty} = 0.8$ km/s case by 0.0472 (4.7 percentage points), but overpredicts the $v_{\infty} = 1.0$ km/s case by 0.0724 (7.2 percentage points). The opposite-sign errors in two neighbouring cells with minimal SPH coverage indicate that the local velocity relationship is

not reliably learned in this region. Therefore, being within the overall sampled input range does not guarantee a reliable prediction, and additional SPH simulations would be useful in such sparsely sampled regions.

3.7 How the Model Supports SPH Decisions

So, what is the model actually useful for? The screening process separates disruption from retention. While BMF calculates the amount of material that is still attached to Mars, the largest-remnant fraction indicates whether the asteroid is largely intact, breaks apart somewhat, or is seriously disturbed. Strong disruption can nevertheless result in primarily fleeing debris, which is why both outcomes are distinct.

The resulting screening workflow provides the breakup class, estimated Mars-bound mass, expected BMF, data coverage, and a recommendation about whether another SPH simulation is required. While sparse, edge, transition, or doubtful instances should be examined with SPH, well-supported cases can be swiftly evaluated with ML.

4. Discussion

The clearest periapsis and velocity trends are seen in the controlled experiments, while the matched comparisons show that the effects of spin and velocity vary with encounter conditions. The behaviour in Figure 3 is consistent with the other analyses: although BMF generally decreases at larger periapsis, Figures 1, 4, and 5 show that retention can still occur under particular combinations of spin, velocity, and mass. This indicates that similar periapsis values can produce different outcomes depending on the full encounter conditions. These relationships are learned from SPH outputs and support interpretation rather than proving physical causality.

The primary physical outcome is that each parameter's impact is contingent upon the other encounter conditions. Within those parameters, periapsis determines the overall strength of the tidal encounter, encounter speed influences the duration of the contact and the favourable capture, and mass and spin alter the outcome. The most obvious periapsis and speed directions are found in the controlled experiments. The paired simulations demonstrate how periapsis and spin state affect the magnitude of the spin and velocity impacts.

Though these associations are acquired from SPH outputs rather than from physical laws, the ML model replicates general trends and reveals interactions throughout the archive. As a result, matched comparisons, controlled tests, and group validation support interpretation but do not prove causality.

As a result, ML's usefulness is restricted. It can swiftly filter well-supported input combinations, identify transition zones, uncover situations that yield promising retained mass, and indicate areas where more simulation could yield valuable information. Long-term debris survivability, precise

fragment shapes and distributions from BMF alone, and forecasts for inadequately sampled input combinations cannot all be determined by it. This is evident from the sparse periapsis-velocity example at 1.4 Mars radii: even though the input values fall within the training ranges, minimal local simulation coverage is associated with larger held-out errors.

Different questions are addressed by the matching and ablation analysis. Although matching cases demonstrate that periapsis or spin can become more significant under specific encounter settings, velocity is generally the strongest.

Compared to BMF, fragmentation requires more prudence. Strong fragmentation is possible but not guaranteed by close periapsis, and the number of pieces is directly altered by the FoF connecting length. Therefore, rather than taking the place of BMF, fragment count and largest-remnant metrics offer additional information. Strong breakup does not always result in much bound debris, therefore the procedure separates breakup severity from Mars-bound retention.

All things considered, the ML model is a quick screening method for the current SPH archive. Poorly sampled cases, transition regions, and issues requiring certain fragment attributes still require SPH.

5. Conclusions, Limitations and Future Work

5.1 Conclusions and Answers to the Research Questions

This study sought to determine which inputs are most important for fragment creation and retained debris, how orbital and physical characteristics affect disruption and bound retention, and whether ML can forecast outcomes while indicating when its predictions should be believed. Below is a summary of the responses.

Question 1: How do orbital and physical parameters affect disruption and bound retention? Although the impact varies depending on the other inputs, closer and slower contacts typically retain more debris. In the representative controlled test, predicted BMF generally decreases from roughly 48% at 1.2 Mars radii to roughly 12% at 3.0 Mars radii. A similarly clear decline occurs as encounter speed increases, from roughly 45% at low velocity to near zero by about 1.2-1.4 km/s. Rather than being universal thresholds, these reactions are condition specific. Mass and spin alter the outcome, speed alters the encounter and capture conditions, and periapsis regulates tidal power.

Question 2: Which inputs matter most for retained debris? There is no single universal ranking under all encounter conditions. For BMF, the tuned Gradient Boosting input-removal analysis identifies velocity as the strongest overall physical input, with its removal causing the largest decrease in performance ($\Delta R^2 = -0.508$), followed by periapsis (-0.466) and spin (-0.246). However, matched cases show that their relative importance changes locally.

Question 3: Can machine-learning models predict outcomes within the sampled data while showing when a prediction should or should not be trusted? Yes, in cases where the SPH archive supports comparable input combinations. The primary tuned Gradient Boosting model using the original scenario inputs obtains a grouped cross-validated R^2 of 0.9328, MAE of 0.0277, and RMSE of 0.04. The tested physics-derived features do not improve the tuned model, so the simpler model using the original scenario inputs is retained. However, performance is not equally dependable everywhere. Despite the input values falling within the numerical training ranges, held-out errors could reach about 7.2 percentage points at periapsis 1.4 Mars radii in a sparsely sampled region. This demonstrates that being within the overall input range is insufficient for a reliable prediction; comparable SPH cases must also exist nearby in the sampled parameter space. Thus, ML can support rapid screening and simulation prioritisation, while sparse, edge, transition, or poorly supported cases still require additional SPH investigation.

Overall, the project shows that tidal-disruption outcomes depend on interacting physical conditions rather than one universal ranking of parameters. Machine learning makes the existing SPH archive faster to explore, shows where predictions are well supported, and highlights where new simulations would be most useful. It complements SPH rather than replacing the simulation physics.

5.2 Limitations

The simulation archive's short size and uneven sampling are its primary drawbacks. A single SPH simulation can be analysed in multiple ways, hence many rows are not independent physical evidence. Although it is unable to produce missing simulations, group validation keeps the test score from becoming unduly optimistic. Even when each individual input value falls between its reported minimum and maximum, predictions are still unclear for combinations that are poorly sampled, close to the data's edges, or outside the combinations observed during training.

Additionally, instead of solving the physical laws directly, the ML model learns patterns from SPH results. It does not follow the subsequent orbits of fragments and does not impose conservation of mass, momentum, energy, or angular momentum. Therefore, a reliable BMF prediction cannot determine whether debris accumulates into moons, forms a disk, or endures for a long time. Additionally, fragment measurements are more sensitive to the FoF connecting length employed in post-processing and are noisier than BMF.

5.3 Future Work

The highest priority for future work is to run new SPH simulations in poorly covered mass-periapsis-velocity-spin combinations, especially near the change between retained and zero BMF. More matched groups should repeat comparison at other masses and resolutions while keeping processing choices fixed. It would be possible to determine whether the stronger spin disparities also occur in other groups by reporting the BMF range together with the actual BMF values and

the number of independent simulations. It would be more robust to test the model on whole new SPH runs created after training rather than repeatedly dividing the current library.

It would be more robust to test the model on whole new SPH runs created after training rather than repeatedly dividing the current library. A practical future workflow would then add validated new SPH outcomes to the training dataset and periodically retrain the surrogate model, allowing newly simulated regions to improve subsequent predictions. Training the tuned Gradient Boosting model takes approximately 7 seconds on the macOS arm64 machine used in this study for the current 407-row training dataset, so repeated retraining would be computationally inexpensive.

Future research should enhance automatic identification of input combinations with inadequate support and per-prediction reliability estimates. Before updating the current model, new SPH simulations should be used to test exploratory models. Stronger physical limitations, multi-output prediction, and uncertainty estimates are possible additional enhancements.

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Appendix A. Supporting Model Results

Table A1. Comparison of BMF regression models using five-fold GroupKFold by physical SPH simulation, including exploratory advanced models to show the potential for further performance improvement in future work.

Model	R ²	MAE	RMSE	Status
Baseline Random Forest (RF)	0.9191	0.0293	0.0439	Baseline model
Tuned RF	0.9318	0.027	0.0403	Tuned model
Tuned RF + derived features	0.9376	0.0268	0.0385	Slightly improved performance
Baseline Gradient Boosting (GB)	0.9262	0.0296	0.0419	Best baseline model
Tuned GB	0.9328	0.0277	0.0400	Main model
Tuned GB + derived features	0.9253	0.0298	0.0421	Slightly reduced performance

The 10% BMF threshold (using logistic regression) is used for screening rather than as a criterion for moon formation, therefore excluded from the table. Tuning Gradient Boosting improves performance to $R^2 = 0.9328$, $MAE = 0.0277$, and $RMSE = 0.0400$. Adding the tested derived features to Gradient Boosting slightly reduces performance. Although Random Forest with derived features achieves slightly higher predictive performance ($R^2 = 0.9376$, $MAE = 0.0268$, $RMSE = 0.0385$), the tuned Gradient Boosting model using the original scenario inputs is retained as the primary model because it achieves comparable performance without requiring additional derived inputs.

Table A2. Model tuning parameters and physics-derived features investigated for gradient boosting model.

Category	Parameters	Baseline	Changed	Performance
Model tuning	Number of trees (n_estimators)	200	500	Increased R ² by 0.066
	Learning rate (learning_rate)	0.1	0.08	
	Subsample	1	0.8	
	Minimum samples per leaf (min_sample_leaf)	2	1	
Simple derived feature	Encounter speed squared	v_∞	v_∞^2	Decreased R ² by 0.001
	Inverse periapsis	r_p	$1/r_p$	
	Spin frequency	P_{spin}	$1/P_{\text{spin}}$	
	Asteroid radius	M_{asteroid}	$R = \left(\frac{3M}{4\pi\rho_{\text{ast}}}\right)^{1/3}$ with $\rho_{\text{ast}} = 2700 \text{ kg/m}^3$	
Combined physics feature	Periapsis-speed interaction	r_p, v_∞	$r_p \times v_\infty$	Decreased R ² by 0.009
	Encounter eccentricity	r_p, v_∞	$e = 1 + \frac{r_p v_\infty^2}{\mu_{\text{Mars}}}$	
	Time within 2 R _{Mars}	r_p, v_∞	$t_{\text{inside}}(r_p, v_\infty; 2R_{\text{Mars}})$	
	Time within tidal-disruption radius	r_p, v_∞	$t_{\text{inside}}(r_p, v_\infty; R_{\text{tidal}})$	

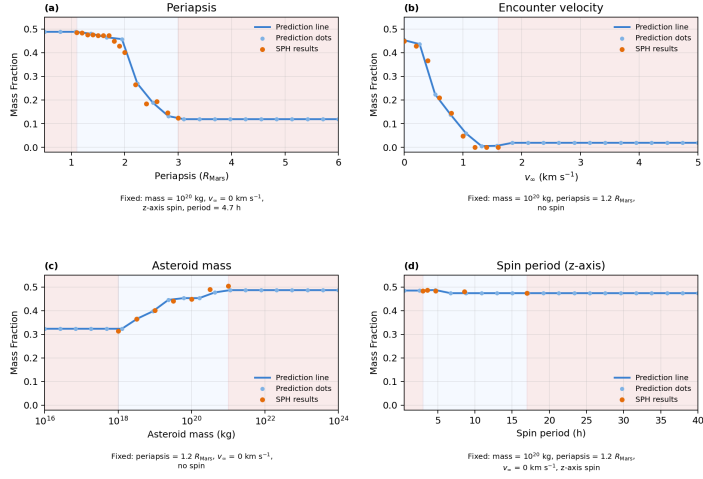


Figure A1. Gradient Boosting predictions beyond the observed SPH parameter ranges.

The shaded central regions indicate the ranges represented by the corresponding SPH results. Outside these ranges, the tree-based model does not extrapolate a physical trend. Instead, it continues to use partitions learned from the training data, producing stepwise or constant predictions, as seen most clearly for periapsis, velocity, and spin. For mass, the prediction keeps increasing slightly beyond the observed range because the higher mass values still cross different learned tree thresholds. Farther outside the range, the values fall into the same tree regions, so the prediction becomes nearly constant. This happens sooner for the other parameter sweeps, where out-of-range values quickly enter the same terminal tree regions and further changes no longer affect the prediction. These out-of-range predictions should therefore not be interpreted as physical trends.

Appendix B. AI acknowledgement Statement

I used OpenAI Codex (<https://chatgpt.com/codex>) throughout the project for targeted code-development support. This included producing initial implementation drafts for data extraction and processing, exploratory data analysis, feature engineering, diagnostic and visualisation scripts, refactoring, debugging, repository documentation, and some commit messages. All AI-assisted code and documentation were reviewed, modified, tested, and verified by me before inclusion. I specified the scientific questions, modelling targets, physics-derived features, thresholds, evaluation strategy, plot selection, interpretation, and final project decisions, and I can explain, modify, and reproduce the submitted implementation. AI was not used to write the Project Plan or Final Report. The scientific conclusions, design decisions, and submitted report are made by me and reflect my own work and understanding. Further details of AI usage are documented in `docs/ai_usage.md`.